

Milan Telecom Traffic Forecasting

This report describes a small empirical forecasting study using the Milan Telecom Traffic 2013 dataset.

Key findings

Model	MAE	RMSE	MAPE
Random Forest	58,206	68,691	50.34%
Linear Regression	61,880	74,288	56.44%
HistGradientBoosting	111,965	123,747	112.08%

Top sampled squares

Rank	Square ID	Total traffic
1	5161	14,614,418
2	5059	13,680,137
3	5259	12,259,581
4	5061	11,228,350
5	6064	10,658,021

Figure: top_areas_by_total_traffic.png

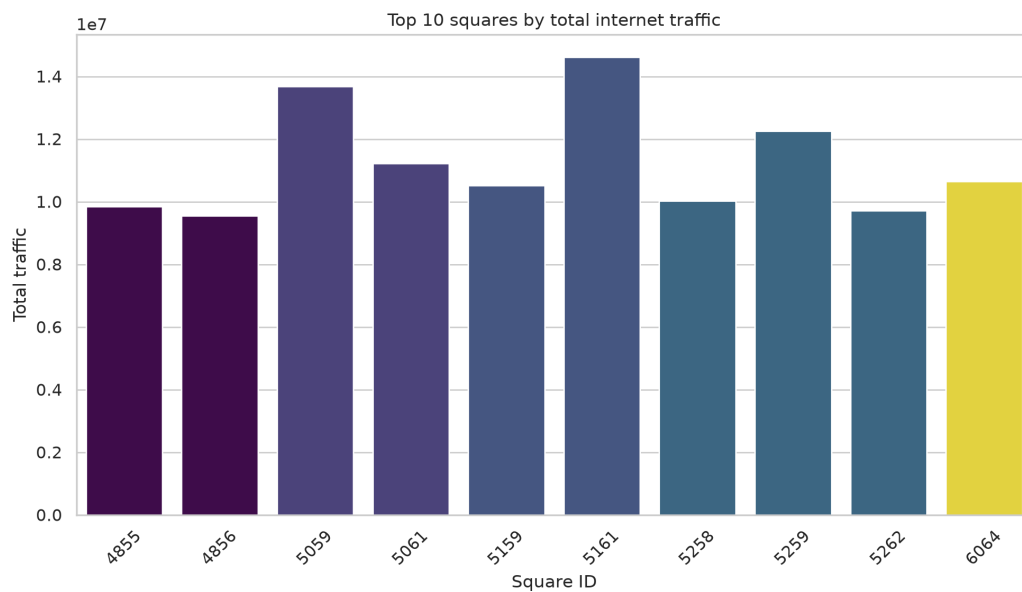
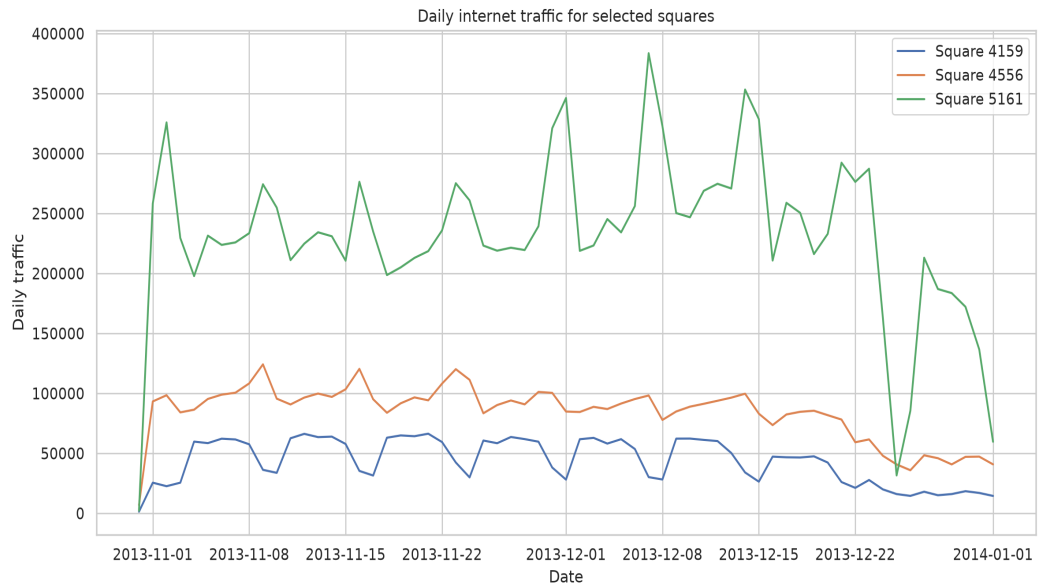


Figure: selected_square_daily_traffic.png



Conclusion

The project finds that a few highly active network cells dominate the overall traffic profile, and that a nonlinear tree-based model provides the strongest baseline for forecasting the busiest square. Future work should expand feature engineering, compare against time-series-specific methods, and evaluate multiple cells in parallel.